

Requirement Analysis

Customer Journey Map — Rajesh (Small-Scale Farmer)

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and

what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Stage	Awareness	Consideration	Onboarding	Usage	Decision / Outcome
Actions	Realizes crop yields have been inconsistent for several years	Hears about a data dashboard showing crop trends for his region	Opens the Tableau Public dashboard for the first time via a shared link	Filters by his state (Uttar Pradesh), season (Kharif), and explores crop production charts	Selects a crop for next season based on 20+ years of historical performance data
Thoughts	"Why do my yields keep changing every year?"	"Could data actually help me plan my crops better?"	"This looks complex — but I can filter by my state and season"	"Rice and wheat have consistently performed well in my region during Kharif"	"I feel confident enough to try a different crop this season"
Emotions	Frustrated, uncertain	Curious, slightly skeptical	Overwhelmed, then reassured	Engaged, empowered	Confident, relieved
Touchpoints	Local farmer network, news, NGO outreach	Government extension officer recommendation	Tableau Public dashboard (shared link)	Interactive filters: State, Season, Crop, Year slider	Final crop and season decision informed by data
Opportunities	Raise awareness of free data tools among rural farmers	Simplify dashboard UI for low-tech users	Add tooltips and guided onboarding walkthrough	Ensure mobile/low-bandwidth compatibility	Track outcomes across seasons; build farmer feedback loop

The emotional curve dips lowest at Onboarding, then climbs steadily as filters (State, Season, Crop) let Rajesh narrow the view to something personally relevant, confirming that usability and guided onboarding are the difference between a farmer abandoning the tool and one who reaches a confident planting decision.